

ENCORB

The Circular Trade Desk

Product & Engineering Build Brief

A phased specification for the Encorb SaaS platform — marketing site, B2B marketplace, transactions, user dashboards, and the automated emissions calculator

Priority 1 · Marketing front end

Goal: a good-looking, credible public website that explains what Encorb is and what it does, and converts visitors into sign-ups. This ships first and can go live before the marketplace is fully wired.

Pages to build

Page	Purpose	Key sections
Home	Explain the platform, build trust, drive sign-up	Hero + live price strip, 3-step flow, EnCore Engine, materials, compliance, CTA
Exchange / How it works	Show the marketplace value before login	Seller walkthrough, buyer walkthrough, sample listings
Materials	Detail the material families traded	Metals, polymers, glass — grades, specs, examples
Security & Compliance	Establish institutional trust	RCRA, ISRI grading, verification, audit trail, SOC 2 (roadmap)
Pricing	Explain plans / fees	Tiered plans or transaction-fee model (OPEN)
About	Company story & mission	Team, mission, circular-economy thesis
Resources	SEO + education hub	Knowledge hub, guides, regulations explainer
FAQ	Reduce sales friction	Common buyer/seller/compliance questions
Contact	Capture leads / demo requests	Form → CRM/inbox

Functional requirements

- Fully responsive (verify at 375px, 768px, 1024px, 1440px); no horizontal scroll on mobile.
- Live price strip / ticker in the header area — reads reference prices for recovered grades (data source wired in Priority 5 groundwork; placeholder feed acceptable for launch, clearly non-binding).
- SEO: server-rendered pages, semantic HTML, meta/OG tags, sitemap, fast Core Web Vitals.
- Accessibility: WCAG 2.1 AA — 4.5:1 contrast, visible focus states, keyboard nav, alt text, prefers-reduced-motion respected.

- Every primary CTA ('List a material', 'Browse the exchange', 'Talk to the desk') routes to the correct next step (marketplace or sign-up).
- Analytics events on all CTAs and key scroll depths.

Priority 2 · The marketplace (listings)

Goal: once the site pages exist, build the marketplace where buyers and sellers come together and browse available listings. This is the core of the product.

Screens

- **Marketplace / listings grid** — the browsable exchange. Cards showing material, grade, tonnage/volume, location, reference price, cadence (one-off vs recurring), and seller verification badge.
- **Filters & search** — by material family (metals / polymers / glass), grade (e.g. Copper #1, HDPE regrind, PET bales), location/region, volume range, price range, recurring vs spot, contamination tolerance.
- **Listing detail page** — full spec sheet, photos/specimen, documentation status, seller profile & verification, reference price and quantity, and the primary action (Buy / Make offer / Contact seller).
- **Create-listing flow (seller side)** — the form a generator uses to post material (behind account creation — see Priority 4).

Data captured per listing

Field	Type	Notes / example
Material family	enum	Metals / Polymers / Glass
Grade	enum/text	Copper #1, Aluminum UBC, HDPE regrind, PET bales, Flint cullet...
ISRI spec code	text	US scrap-grading reference where applicable
Volume / tonnage	number + unit	e.g. 42 t/mo
Cadence	enum	One-off (spot) or recurring stream
Contamination tolerance	number/%	e.g. <2%
Location	text/geo	City, state — drives logistics & emissions
Reference price	number (USD)	Per unit; may be indicative
Documentation	files	Weigh tickets, BOL, diversion cert, spec sheet
Photos / specimen	images	Material condition
Seller / org	ref	Verified company (KYC/EIN)
Status	enum	Draft / Live / Under offer / Closed

Access model: Browsing listings can be public or lightly gated (OPEN — decide whether anonymous visitors see full detail or a teaser). Acting on a listing always requires a verified account (Priority 4).

Note on the 'live order book' promise

The design shows a live, liquid market feel (price strip, match panel, order book). That creates a real product obligation: the exchange must have enough genuine listings at launch to make this credible. Please build the listings system so it degrades gracefully with low volume (e.g. 'new listings arriving' states) rather than showing an empty book.

Priority 3 · Transactions

Goal: once a user finds a material they want, they can act on it — initiate a transaction (buy, sell, or make an offer) directly from the listing.

Transaction flow

- From a listing detail page, the buyer clicks the primary action (Buy / Make offer).
- If not logged in, they are prompted to create an account / sign in (Priority 4) — the account gate sits here.
- Once authenticated & verified, they can submit an offer or accept the listed terms.
- The seller is notified, reviews, and accepts / counters / declines.
- On acceptance, a deal record is created linking buyer, seller, listing, agreed price, and quantity.
- Deal moves through states: Offer → Accepted → In fulfilment → Settled → Closed.
- **On settlement, the emissions calculation fires automatically (Priority 5).**

What a deal record stores

- Buyer org, seller org, and the individual users who acted.
- Listing reference + snapshot of agreed spec (grade, volume, contamination).
- Agreed price (USD), quantity, and agreed logistics (mode of transport, origin, destination).
- Documents: BOL, weigh tickets, diversion certificate, chain-of-custody / audit trail.
- Timestamps for each state transition.
- Emissions result once calculated (see Priority 5).

Open decisions for v1

Question	Options	Recommendation
Payment/settlement	In-platform escrow vs off-platform (invoice)	Off-platform for v1; record terms only. Escrow later.
Offer model	Fixed-price 'Buy now' vs negotiated offers vs auction	Support Buy-now + Make-offer at launch; auctions later.
KYC depth	Light (EIN + email) vs full docs	Light at launch, escalate before settlement.

Question	Options	Recommendation
Contracts	Generated agreement vs external	Generate a simple deal summary PDF; e-sign later.

Priority 4 · Accounts & user dashboard

Goal: when a user wants to buy or sell, they are prompted to create an account using their company email. Each user gets a dashboard showing their details, deals done, deals open, and more.

Account creation

- **Trigger:** the account modal/pop-up appears the moment a user tries to buy or sell (i.e. at the transaction gate), not before browsing.
- **Company email required:** sign-up must use a company email address. Block free/consumer domains (gmail, yahoo, outlook, etc.); verify the domain via a confirmation link.
- Email verification (magic link or code) before the account is active.
- Org model: the email domain maps to a company/organization. First user from a domain can create the org; later users from the same domain join it (with an approval step). This lets a company have multiple users under one account.
- Capture at sign-up: name, role, company name, company email, phone (optional), and whether they are primarily a buyer, seller, or both.
- Verification / KYC: collect EIN and business details; verification status gates the ability to transact and to settle.

User dashboard — sections

Section	What it shows
Overview	Snapshot: open deals, deals in progress, closed deals, total volume traded, total emissions avoided/associated.
My deals	All deals with status filter — Open / In progress / Settled / Closed. Each links to the deal detail & documents.
My listings (sellers)	Their posted materials, status (Live / Under offer / Closed), edit & re-list.
Offers / activity	Incoming and outgoing offers awaiting action.
Emissions & reporting	Per-deal and aggregate emissions from the calculator; export for Scope 3 / ESG reporting.
Documents	BOLs, weigh tickets, diversion certs, audit trail per deal.
Company profile	Org details, verification status, team members, KYC/EIN status.
Account settings	Personal details, notifications, security.

Roles & permissions

- Org Admin: manages company profile, approves new users on the domain, sees all org deals.
- Member: creates listings / makes offers, sees their own and (configurably) the org's deals.
- Verification status (Unverified / Verified) is a gate on transacting and settling.

Priority 5 · Emissions calculator

Goal: an automated calculator — logic first prototyped as an Excel sheet — that computes the emissions associated with a completed deal. It runs automatically once a deal is done between buyer and seller.

When it runs

The calculation fires automatically on deal settlement (Priority 3). The result is stored on the deal record and surfaced in both the buyer's and seller's dashboards, and is exportable for Scope 3 / ESG reporting.

Inputs (drivers of the calculation)

Input	Source	Why it matters
Mode of transport	Deal / logistics	Truck, rail, sea, etc. — each has a different emission factor per ton-mile.
Type of waste	Listing	Waste stream category affects factors and diversion credit.
Type of material	Listing	Metals / polymers / glass — recovery vs virgin-production offset differs by material.
Buyer location	Buyer org	Destination point for distance calculation.
Seller location	Seller org / listing	Origin point for distance calculation.
Quantity / tonnage	Deal	Scales the whole calculation.

Calculation logic (from the Excel model)

- Distance between seller and buyer is derived from their locations (geocode origin/destination → distance).
- Transport emissions = distance × quantity × emission factor for the chosen mode of transport (per ton-mile).
- Material/processing factor applied based on material and waste type.
- Diversion / avoided-emissions credit applied where recovered material displaces virgin production (material-dependent).
- Net result = associated emissions (and, where modelled, emissions avoided) for the deal, in kg or tonnes CO2e.

The existing Excel calculator defines the exact formulas, factors, and lookup tables. **It should be treated as the specification for this module** and ported into the backend so it runs server-side per deal, not maintained as a manual spreadsheet.

Reference data / factors

Emission factors and material data should come from recognised US sources so the numbers are defensible for ESG reporting. Candidates already identified: EPA WARM (material-specific), plus transport factors from

standard freight sources. Pricing/market data sources under consideration include LME, Fastmarkets, AMM, RecyclingMarkets.net, ReMA and USGS.

Requirements

- Port the Excel formulas into a server-side calculation service with the factor tables in the database (versioned, so factors can be updated without code changes).
- Store the full input snapshot with each result, so a calculation is reproducible and auditable.
- Expose the result per deal and in aggregate on dashboards; provide CSV/Excel export for reporting.
- Keep the Excel sheet as the canonical reference during build so outputs can be validated against it row-for-row.